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Operations and Business Growth
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Inventory Management: EOQ, Safety Stock, and Working Capital

Summary: Balance stockout risks against expensive storage and capital lockup. Compute Economic Order Quantity (EOQ) and Reorder Points.

Learning Objectives

- Distinguish between ordering costs and holding/carrying costs.
- Calculate Economic Order Quantity (EOQ).
- Establish Reorder Points (ROP) with safety stock buffers.
- Prevent dead inventory while avoiding revenue-destroying stockouts.

Core Principles & Key Takeaways

- EOQ minimizes the total cost of ordering and carrying stock.
- Safety stock protects against unpredictable supplier lead times and sudden demand spikes.
- Audit slow-moving inventory regularly to free trapped cash.

1. The Inventory Dilemma: Holding Costs vs. Stockouts

Inventory represents cash sitting on shelves. Excessive inventory ties up working capital, incurs warehouse rent, risks spoilage or obsolescence, and incurs holding costs. In contrast, insufficient inventory causes stockouts, lost sales, and damaged customer trust.

2. Economic Order Quantity (EOQ) Formula

The **EOQ** identifies the optimal order batch size that minimizes the sum of inventory ordering costs and carrying costs:

$$\text{EOQ} = \sqrt{\frac{2 \times D \times S}{H}}$$

Where:
* **D:** Annual customer demand in units.
* **S:** Fixed order placement cost (freight, inspection, bank transfer charges per order).
* **H:** Annual holding/carrying cost per unit (rent, insurance, shrinkage, financing interest).

3. Reorder Point (ROP) with Safety Stock

$$\text{Reorder Point (ROP)} = (\text{Daily Demand Rate} \times \text{Lead Time in Days}) + \text{Safety Stock}$$

If average lead time to receive stock from port to store is 14 days, and daily sales are 50 units, with a 200-unit safety stock buffer:

$$\text{ROP} = (50 \times 14) + 200 = 700 + 200 = 900 \text{ units}$$
When warehouse stock reaches 900 units, the inventory system triggers a purchase order.

Self-Assessment & Review Questions

Q1: What happens if a company doubles its order batch size without an increase in customer demand?

A. Holding costs increase, while annual order placement frequency decreases
B. Total inventory cost is guaranteed to drop
C. Safety stock becomes zero
D. Cash flow immediately improves

Correct Answer: A • Larger order batches mean fewer annual orders (saving ordering cost), but higher average warehouse inventory (increasing annual holding costs).

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